Foundry (Heavy Industry / Close-Range Ground Combat & Destruction)

Executive Summary

Foundry is a brutal **heavy-industry battlefield** built for **close-range ground combat, destructive engagements, interior firefights, and autonomous maneuvering through dense industrial clutter**. The terrain features steel mills, blast furnaces, conveyor tunnels, slag pits, loading bays, pipe corridors, molten-metal hazards, and tight industrial choke zones. The map emphasizes Obsidian Protocol's autonomous interior-navigation logic, sensor-degraded combat, comms-relay warfare, and logistics under extreme industrial density. In AR (Nreal Light/Air), the map anchors at ~4–5 m viewing distance, covering a 10x10 m physical area representing ~1 km² of heavy-industry terrain. Key features include furnace-side combat, pipe-maze infiltration, slag-pit hazards, and destructive close-quarters engagements.

Design Goals & Gameplay

Close-Range Industrial Combat

Foundry prioritizes brutal interior firefights, short sightlines, ambushes, and destructive engagements. Heavy machinery creates unpredictable cover and hazards.

Destruction-Driven Gameplay

Pipes rupture, slag pits ignite, conveyor belts jam, and machinery collapses. Environmental destruction reshapes routes and cover dynamically.

Autonomy & Interior Pathfinding

Autonomous units choose between pipe corridors, furnace catwalks, loading-bay flanks, or slag-pit bypasses. Units adapt to blocked machinery and shifting hazards.

Sensor & Information Warfare

Heat shimmer, smoke, sparks, steam, and metal clutter distort sensors. Relay towers on gantries restore clarity. Destroying relays creates blind pockets across the foundry.

Logistics & Resources

Fuel tanks, maintenance bays, and loading docks serve as resupply nodes. Convoys must navigate narrow industrial lanes and avoid ambush-prone machinery corridors.

Deployment & Progress

A fixed deployment budget (~10,000 points) forces players to choose between heavy armor, demolition units, or autonomous infantry. Blue deploys from the southwest loading bay; Red deploys from the northeast furnace control tower.

Spatial Constraints (Nreal AR)

Device Capabilities

Nreal Light/Air: ~53° diagonal FOV, 1920×1080 per-eye resolution. Optimal viewing distance ~4–5 m.

Playable Area & Scaling

10×10 m physical area ≈ 1 km² industrial zone (1:100 scale). Designed for tabletop AR.

Tracking & Occlusion

Inside-out tracking detects horizontal planes. Depth-mesh (if available) provides limited occlusion. Virtual objects render over real-world view; assume minimal real-world occlusion.

Layout Overview

Coordinate System

1000×1000 grid (1 unit ≈ 1 m). Origin (0,0) = southwest corner.

Key Locations

Blue HQ (Loading Bay Command) — (0,0), 150×150 m Cargo loaders, forklifts, armored staging area.
Red HQ (Furnace Control Tower) — (1000,1000), 150×150 m High-heat command node with radar mast and AA emplacements.
Central Foundry Core — (500,450), 300×300 m Blast furnaces, molten-metal channels, catwalks, heavy machinery.
Maintenance Depot — (250,800), 120×120 m Tools, generators, repair bays, fuel drums.
Pipe-Maze Corridor — (800,300), 200×200 m Dense pipe networks, steam vents, interior flanking routes.
Slag Pit Zone — (0,450 → 400,900) Hazardous chokepoint: molten slag, collapsing platforms.
Conveyor Route — (300,0 → 450,350) Moving belts, machinery cover, narrow traversal.
Gantry Relay A — (650,150), +240 m Long-range sensor vantage point.
Rooftop Relay B — (150,650), +260 m Comms relay tower and recon hub.
Neutral Forward Outpost — (500,950) Drone resupply zone.

Chokepoints & Sightlines

Slag Pit Zone

Hazardous chokepoint; environmental destruction reshapes combat.

Pipe-Maze Corridor

Interior flanking routes; extremely tight sightlines.

Foundry Core

Heavy machinery creates dynamic cover; ideal for demolition units.

Conveyor Route

Predictable movement path; vulnerable to ambushes.

Sightlines

Gantry Relay A provides long-range sensor coverage (~300 m). Heat shimmer and smoke create fog-of-war pockets.

Deployment Zones

Blue Deployment: SW loading bay + armored staging.
Red Deployment: NE furnace control tower + AA emplacements.
Neutral Forward Zone: Outpost at (500,950) for drone drops.

Assets & Environment

Terrain

Furnaces, slag pits, conveyor belts, pipe corridors, loading bays.

Vegetation

None — industrial debris only.

Buildings & Props

Military: AA guns, radar dishes, watchtowers. Industrial: Furnaces, cranes, pipelines, generators. Logistics: Trucks, crates, fuel drums, drones.

Effects

Smoke, sparks, dust, heat shimmer, steam vents, fire bursts.

Audio

Machinery, metal impacts, steam, radio chatter.

LOD & Budget

Asset Cat	Count	Tri/Item	Total Tris
Furnaces/	3	200k	600k
Pipes/Ind	4	80k	320k
Buildings/	15	50k	750k
Roads/Pa	4	30k	120k
Props	50	10k	500k
Total	—	—	**~2.29M**

Texture Budget

2048² for terrain/buildings, 1024² for props. DXT1/5 compression. Target ≤100 MB.

Unreal / Nreal Integration

Engine & Plugins

Unreal ARTemplate + Nreal SDK (XREAL). ARKit/ARCore enabled.

Rendering

Forward renderer, stationary lights, baked GI, aggressive culling.

Anchors

Single spatial anchor for entire map.

Interaction

Phone pointer for pan/zoom; HUD as screen-space widgets.

Multiplayer

Client-server architecture; offload AI/physics to server.

Implementation Plan & Testing

Milestones

1. **Prototype Layout — Month 1** Block-out furnaces, pipe maze, slag pits, AR anchor test.
2. **Core Systems — Month 2** NavMesh, interior autonomy behaviors, destruction logic.
3. **AR Integration — Month 3** Anchors, plane detection, basic interactivity.
4. **Content Fill — Month 4–5** Final models, textures, lighting, audio.
5. **Gameplay & Polish — Month 6–7** Objectives, balance, VFX (sparks, heat shimmer, steam).
6. **Testing & QA — Month 8** Multiplayer stress test, sensor validation.

Testing Plan

Autonomy

Verify interior navigation, destruction-aware pathfinding, hazard avoidance.

Comms/Sensors

Test relay destruction, fog-of-war updates, LOS checks.

Performance

≥60 Hz AR render on target device.

QA Checklist

Navigation, destruction logic, deployment budget enforcement, anchor drift, sensor accuracy.